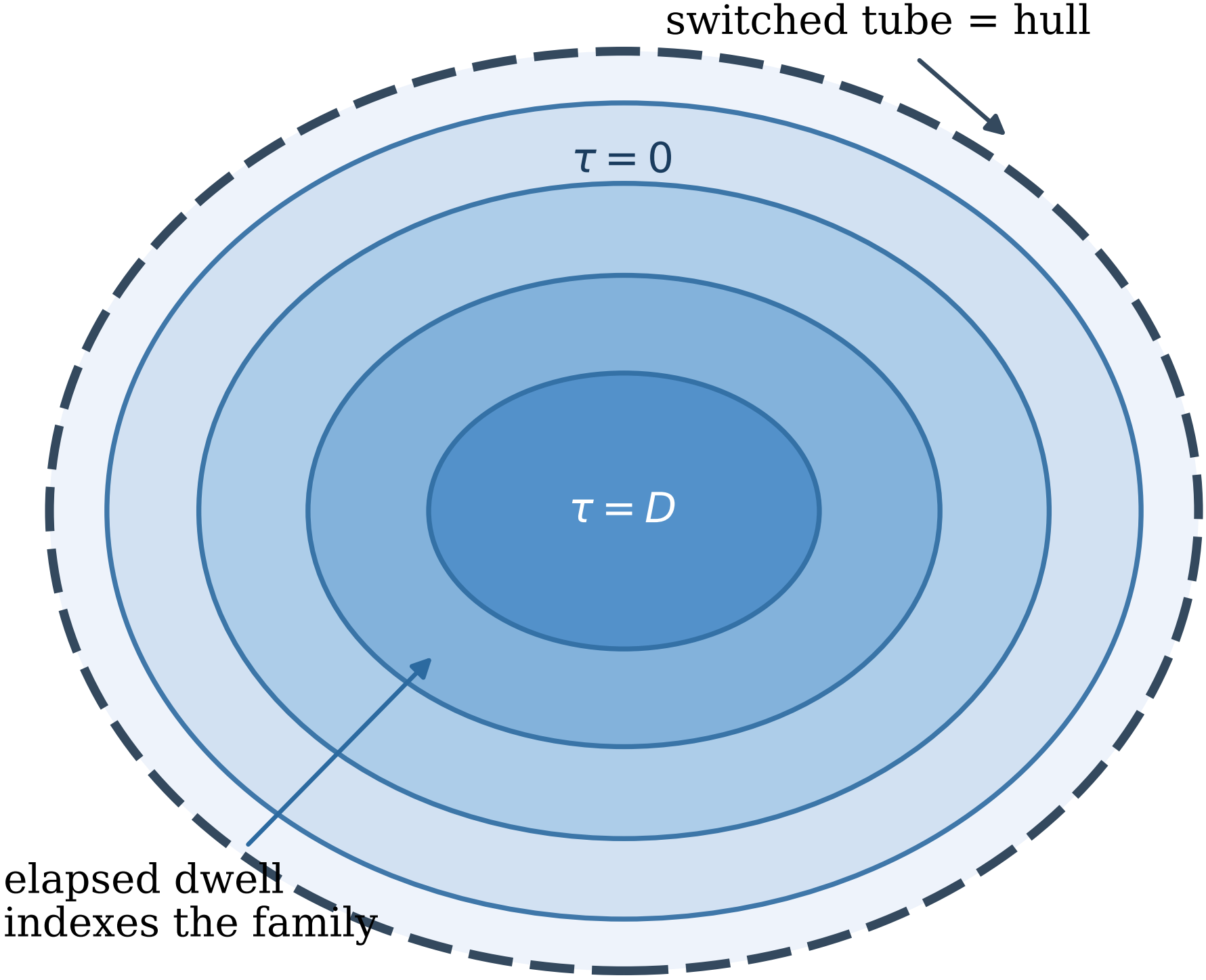
switched tube = hull
$\tau = 0$
$\tau = D$
elapsed dwell
indexes the family